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Association Between Socio-Demographic Characteristics and Household Food Waste Behaviors Among University Students in Italy: The UniFoodWaste Study

Lorenzo Stacchini [1], Daniele Nucci [2], Nicoletta Romano [2], Maria Villa [2], Chiara Passeri [2], Roberta Ferranti [2] and Vincenza Gianfredi [3,*]

[1] Department of Health Sciences, University of Florence, 50121 Florence, Italy
[2] Struttura Semplice Dipartimentale Igiene Alimenti e Nutrizione, Dipartimento di Igiene e Preven-Zione Sanitaria, Agenzia di Tutela della Salute (ATS) Brescia, Via Duca Degli Abruzzi, 15, 25124 Brescia, Italy
[3] Department of Biomedical Sciences for Health, University of Milan, Via Pascal, 36, 20133 Milan, Italy
* Correspondence: vincenza.gianfredi@unimi.it

Abstract: Food waste is a growing environmental, economic, and social challenge worldwide. This study aimed to assess the association between various socio-demographic characteristics and household food waste behaviors among university students in Italy. A cross-sectional study was conducted using a validated, online, anonymous, and voluntary-based questionnaire distributed among university students in Milan, Italy. Linear regression analyses were performed to explore the association between socio-demographic factors and anti-household food waste behaviors (preventive practice, abilities scores, and parental guidance). A total of 2779 students participated in the survey. Students residing outside Lombardy exhibited higher preventive practice scores ($\beta = 0.11$, $p = 0.030$). Regular smoking was negatively associated with preventive practice ($\beta = -0.22$, $p < 0.001$) and abilities scores ($\beta = -0.11$, $p = 0.009$). Cohabitation with a partner positively impacted abilities scores ($\beta = 0.23$, $p = 0.007$). Larger family sizes and students in scientific fields showed higher scores for parents' guidance ($\beta = 0.18$, $p < 0.001$). Frequent use of food delivery apps correlated with lower scores across all categories, while anti-food waste apps had a positive effect. Socio-demographic factors significantly influence food waste behaviors among university students. Tailored public health policies and educational programs addressing these factors could effectively reduce food waste.

Keywords: food waste; household behavior; university students; socio-demographic factors; Italy; public health

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1. Introduction

On a global scale, approximately one-third of all food produced for human consumption is lost or wasted, amounting to about 1.3 billion tons per year [1]. This staggering amount includes significant losses at the household level, which vary widely between countries and regions. Recent estimates suggest that Italian households waste approximately 524.1 g per capita per week [2–4]. Food waste represents a significant multifaceted global challenge, impacting environmental sustainability, economic efficiency, and social equity [5]. When considering the environmental consequences, food waste significantly contributes to greenhouse gas emissions, water wastage, and the unnecessary consumption of agricultural resources [6]. Globally, food waste accounts for approximately 8% of annual greenhouse gas emissions. This not only exacerbates climate change but also depletes precious environmental resources, such as water and arable land [7]. At the same time, the economic implications of food waste are profound, with the global cost estimated at approximately $940 billion per year [8]. This staggering figure represents not just the value of the wasted food itself but also the wasted investment in agriculture, transportation, and retail sectors. The issue of food waste also has important social equity dimensions [9].

While significant quantities of food are wasted, food insecurity and malnutrition remain pressing challenges in many parts of the world, including within developed countries like Italy. This dichotomy highlights a significant social injustice, where surplus food coexists with widespread hunger.

In recent years, attention has increasingly focused on understanding and mitigating food waste at the household level, recognizing its contribution to the overall waste stream. Household food waste specifically refers to food discarded in the domestic setting, whether during meal preparation or from uneaten leftovers [10]. This type of waste is particularly concerning because it represents the final link in the food supply chain, where the prevention of waste can have immediate environmental, economic, and social benefits [11]. Unlike food loss, which occurs at earlier stages of the supply chain due to reasons such as poor infrastructure or logistics, household food waste is often the result of consumer behaviors, including over-purchasing, poor planning, and misunderstanding of food labels [12].

A growing body of research has explored food waste behaviors within households and the general population, but there is a notable gap in understanding the factors that influence food waste among specific demographic groups, such as university students. Studies in various countries have indicated that college students may exhibit distinct food waste patterns compared to the general population, largely due to their unique living conditions, financial constraints, and limited cooking experience [13]. University students often live independently for the first time, making decisions about food purchasing and preparation without the guidance they might have received at home. These unique living conditions and lifestyle choices can significantly impact their food waste patterns. Research conducted in the United States and the United Kingdom suggests that factors such as irregular eating habits, lack of food storage knowledge, and time constraints contribute significantly to food waste among students [14]. Moreover, studies have shown that students often face challenges in food management, including planning and budgeting, which can lead to the disposal of edible food [15].

While several studies have explored food waste in households and among the general population, there remains a significant gap in understanding the specific behaviors and influences among university students in Italy. Given the significant role that socio-demographic characteristics—such as age, gender, and living arrangements—play in shaping students' food-related behaviors, it is essential to explore how these factors influence food waste among this group. Additionally, the use of digital tools, such as food-related apps, may further affect students' food purchasing and consumption habits, thus impacting food waste generation.

This study aims to bridge this gap by assessing how socio-demographic characteristics such as age, sex, residency, educational level, and use of food apps, among others, relate to household food waste behavior in this population. Such insights are critical for developing tailored interventions that can effectively reduce food waste among university students. The primary objective of this study is to measure household food waste behavior, and secondly, to assess the association between socio-demographic characteristics and household food waste behavior among university students in Italy. By identifying key factors influencing waste behavior, this research seeks to contribute to the development of more effective public health policies and educational programs aimed at reducing food waste. This, in turn, can lead to significant improvements in environmental sustainability, economic efficiency, and social equity.

2. Materials and Methods

2.1. Study Design

This research employed a cross-sectional study design to explore the relationship between various socio-demographic characteristics and household food waste behaviors among university students in Italy. Data were collected between July and October 2023, allowing for a comprehensive snapshot of the study period. The cross-sectional approach enabled the researchers to analyze data from a large number of participants at a single point

in time, thereby facilitating the identification of patterns and correlations. More details have been previously published [16], providing additional context and validation for the research design.

2.2. Participants

Eligibility for participation in the study was determined based on the following criteria: being a registered university student at University of Milan, aged 18 years or older, residing in Italy during the study period, and willing to provide informed consent. Participation was entirely voluntary, with no incentives provided for completion or participation. This ensured that the data collected reflected genuine responses without any external influences. The participant pool was intended to represent the broader demographic diversity of university students, capturing a range of ages, backgrounds, and living situations.

2.3. Data Collection

Data were anonymously collected using a self-administered, online questionnaire distributed through university email lists. The questionnaire included sections on socio-demographic characteristics, food waste behavior, and the use of food apps for delivery and waste prevention. The questionnaire was implemented using Microsoft Forms (Microsoft 365 A5) and distributed via the university's official mailing list to ensure that only University of Milan students could participate. To guarantee that only University of Milan students could participate, access to the questionnaire was restricted to university users, and measures were taken to prevent duplicate submissions. Additionally, to tackle the problem of incomplete responses or missing data, all questions were set as mandatory. This requirement prevented users from moving to other sections of the questionnaire until they had answered all questions, thereby improving the completeness and reliability of the collected data. Before filling out the questionnaire, participants were asked to review the study's objectives, along with guidance on how to participate. Additionally, each participant was required to give informed consent, and those who chose not to consent were directed to the end of the questionnaire.

2.4. Socio-Demographic Data

The initial section of the survey delved into socio-demographic aspects such as gender (options: female, male, prefer not to say), age, region of residency, smoking habit (yes/no), level of education (high school, degree or higher), cohabitation (alone, with family, with a partner, with friends/roommates), family members, student status (commuter, off-site, living in Milan), study area (health sciences, humanities, science and technology, social sciences), and the use of apps related to food (including delivery and saving food). This comprehensive demographic profiling allowed for a nuanced understanding of the participants' backgrounds and potential influences on their food waste behaviors.

2.5. Household Food Waste

The household food waste behaviors (HFWB) questionnaire was used to gather data on food waste behaviors, including precautionary measures, skills related to food management, and parental guidance on waste prevention [17]. In detail, the HFWB questionnaire is a validated and widely utilized tool in research, featuring 10 questions on a Likert scale that span from 0 (indicating never or strong disagreement) to 6 (signifying always or strong agreement). In the study, its objective was to gauge consumer practices concerning food wastage. Specifically, the questionnaire aimed to measure (i) preventive practices (such as strategizing for grocery shopping and food utilization, eschewing impulsive buys, keeping track of food inventory, preparing meals in suitable amounts, and repurposing leftovers), (ii) abilities (such as perceived difficulties in determining food freshness, inventive cooking, precise planning, and knowledge about prolonging food life), and (iii) the influence of parental guidance on minimizing food waste (highlighting parental stress on reducing food wastage). An overall total food waste score was calculated as the mean between the

scores of the three sections (preventive practices, abilities, influence of parental guidance on minimizing food waste). The final score ranged between 0 and 6, providing a quantifiable measure of household food waste behavior.

2.6. Statistical Analysis

Descriptive statistics were used to summarize the socio-demographic characteristics of the participants and their food waste behaviors, reporting the mean and standard deviation (SD) for continuous variables and the frequency and percentages for categorical ones. Multicollinearity between each predictor was assessed using the Pearson's correlation test ($|\,\text{rho}\,| \geq 0.7$). We calculated the score of each component as the mean of the items of each component and the overall waste score as the mean of the items of the HFWB questionnaire. Due to the continuous nature of each score, we decided to perform four linear regression models (three models for the three component scores and one model for the overall waste score) to assess the association between each socio-demographic factor and anti-food waste behavior, adjusting for age and sex. All the scores were standardized before performing linear regression models. The beta coefficients (β) and 95% confidence intervals (CIs) were calculated to estimate the strength and direction of these associations. A *p*-value of less than 0.05 was considered statistically significant. All statistical analyses were performed using R version 4.3.3, ensuring robust and reproducible results.

2.7. Sample Size Calculation

The calculation of the sample size was performed using a confidence level of 95% and a margin of error of 5%. The reference population for the study was the total student population enrolled at the University of Milan during the 2021/2022 academic year, which consisted of 60,988 students (53). Since the proportion of the Italian student population exhibiting food waste behaviors is not known, we adopted a conservative approach, assuming a proportion value of 50%. Based on these parameters, the minimum required sample size was calculated to be 382 participants.

2.8. Ethical Approval

This study received ethical approval from the Ethical Committee of the University of Milan, ID n. 71.23. This approval underscores the study's adherence to ethical standards in research, ensuring that participants' rights and welfare were protected throughout the study.

3. Results

3.1. Participant Demographics

The study initially surveyed 2779 university students; however, 82 students withdrew consent for data processing and were therefore excluded, resulting in a final sample size of 2697 subjects. Participants were nearly equally distributed among the age groups, with 52.4% (*n* = 1414) aged between 18 and 23 years, and 47.6% (*n* = 1283) aged 24 years or older. The majority of respondents were female (70.5%, *n* = 1902), non-smokers (71.4%, *n* = 1927), held high school diplomas (52.5%, *n* = 1417), lived with their own families (68.4%, *n* = 1845), and were commuters (51.8%, *n* = 1398). Academically, the students were distributed across several fields, with 35.0% (*n* = 944) studying Science and Technology, followed by 24.1% (*n* = 649) in Humanities, 22.8% (*n* = 616) in Health Sciences, and 18.1% (*n* = 488) in Social Sciences.

Regarding food apps usage, 57.4% (*n* = 1547) reported not using food delivery apps, 38.9% (*n* = 1050) used them rarely, and only 3.7% (*n* = 100) used them at least once a week. Similarly, most students did not use food waste apps (63.5%, *n* = 1712), while 33.9% (*n* = 914) used them infrequently, and a small fraction (2.6%, *n* = 71) reported weekly usage.

The students' self-reported scores on prevention practices averaged 5.51 (SD = 0.75), while abilities scored 5.03 (SD = 0.97). Education from parents received a higher rating, with an average score of 5.91 (SD = 1.18), and the overall food waste management score

was 5.27 (SD = 0.76). These results indicate a strong influence of parental education on personal habits and a moderate engagement with health and technology among the student population. Results are reported in Table 1.

Table 1. Descriptive characteristics of the sample.

Characteristic	*n* = 2697
Gender	
Female	1902 (70.5%)
Male	751 (27.8%)
Non-binary	44 (1.6%)
Age	
18–23	1414 (52.4%)
24+	1283 (47.6%)
Region	
Lombardy	2180 (80.8%)
Other	517 (19.2%)
Smoking habit	
Not Smoker	1927 (71.4%)
Smoker	770 (28.6%)
Level of education	
High School degree	1417 (52.5%)
Bachelor's degree or more	1280 (47.5%)
Cohabitation	
Alone	230 (8.5%)
Friend or Roommate	296 (11.0%)
Own family	1845 (68.4%)
Partner	326 (12.1%)
Family Members	
1–3	1074 (39.8%)
4+	1623 (60.2%)
Student status	
Resident in Milan	698 (25.9%)
Off-site	601 (22.3%)
Commuter	1398 (51.8%)
Studying area	
Health Science Studies	616 (22.8%)
Science and Technology Studies	944 (35.0%)
Social Studies	488 (18.1%)
Humanities Studies	649 (24.1%)
Food Delivery App	
No	1547 (57.4%)
Rarely	1050 (38.9%)
At least once a week	100 (3.7%)
Food Waste App	
No	1712 (63.5%)
Rarely	914 (33.9%)
At least once a week	71 (2.6%)
Prevention Practices Score (mean and SD)	5.51 (0.75)
Abilities Score (mean and SD)	5.03 (0.97)
Education from Parents Score (mean and SD)	5.91 (1.18)
Food waste total score (mean and SD)	5.37 (0.71)

n (%); Mean (SD).

3.2. Association Between Socio-Demographic Characteristics and Food Waste

No collinearity was found between predictors (Figure S1).

The linear regression analysis of various predictors on preventive practice, abilities, parents' guidance (Figure S2), and total food waste (Figure S3) scores among university students yielded several noteworthy findings (Table 2). Residents outside Lombardy had a slightly higher preventive practice score, with a beta coefficient of 0.11 (95% CI: 0.01; 0.20, $p = 0.030$). Regular smoking was negatively associated with preventive practice and abilities scores, showing significant reductions in scores [$\beta = -0.22$ (-0.31; -0.12), $p < 0.001$ and $\beta = -0.11$ (-0.19; -0.03), $p = 0.009$, respectively]. Importantly, smokers were also significantly associated with a reduction in total food waste scores ($\beta = -0.20$ (-0.28; -0.11), $p < 0.001$).

Table 2. Multivariate linear regression (based on 2697 subjects) adjusted by age and sex, assessing the association between socio-demographic characteristics and food waste scores (stratified by HFWB questionnaire sections, namely preventive practice, abilities, and parents' guidance; and total score). In bold are the reported statistically significant associations. Statistical significance was set at $p \leq 0.05$.

	Preventive Practice Score	**Abilities Score**	**Parents' Guidance Score**	**Total Food Waste Score**
	Beta (95% CI) *p*-value	Beta (95% CI) *p*-value	Beta (95% CI) *p*-value	Beta (95% CI) *p*-value
Region				
Lombardy	(ref.)	(ref.)	(ref.)	(ref.)
Other	**0.11 (0.01; 0.20) 0.030**	−0.02 (−0.12; 0.07) 0.673	−0.06 (−0.15; 0.04) 0.241	0.05 (−0.05; 0.14) 0.345
Smoking habit (yes)	**−0.22 (−0.31; −0.14) < 0.001**	**−0.11 (−0.19; −0.03) 0.009**	−0.06 (−0.15; 0.02) 0.142	**−0.20 (−0.28; −0.11) < 0.001**
Level of education				
High School diploma	(ref.)	(ref.)	(ref.)	(ref.)
Bachelor's degree or more	0.06 (−0.03; 0.15) 0.173	**0.07 (−0.02; 0.16) 0.119**	−0.05 (−0.14; 0.04) 0.315	0.07 (−0.02; 0.16) 0.146
Cohabitation				
Alone	(ref.)	(ref.)	(ref.)	(ref.)
Friends/roommate	0.11 (−0.06; 0.29) 0.194	0.06 (−0.11; 0.23) 0.508	0.02 (−0.15; 0.20) 0.787	0.10 (−0.07; 0.27) 0.255
Own family	−0.02 (−0.16; 0.12) 0.784	0.07 (−0.07; 0.21) 0.333	**0.18 (0.04; 0.32) 0.010**	0.05 (−0.09; 0.19) 0.521
Partner	0.10 (−0.07; 0.27) 0.244	**0.23 (0.06; 0.40) 0.007**	−0.14 (−0.31; 0.03) 0.098	0.16 (−0.01; 0.32) 0.069
Family member				
1–3	(ref.)	(ref.)	(ref.)	(ref.)
≥4	−0.05 (−0.13; 0.03) 0.221	−0.05 (−0.13; 0.03) 0.225	**0.18 (0.10; 0.25) <0.001**	−0.03 (−0.11; 0.05) 0.440
Student status				
Resident in Milan	(ref.)	(ref.)	(ref.)	(ref.)
Off-site	0.10 (−0.01; 0.21) 0.073	0.01 (−0.09; 0.12) 0.808	**−0.13 (−0.24; −0.02) 0.023**	−0.05 (−0.06; 0.16) 0.369
Commuter	−0.02 (−0.12; 0.07) 0.595	0.00 (−0.10; 0.09) 0.921	−0.03 (−0.12; 0.06) 0.476	−0.02 (−0.11; 0.07) 0.646
Study area				
Health sciences	(ref.)	(ref.)	(ref.)	(ref.)
Science and technology	0.08 (−0.02; 0.18) 0.119	**0.15 (0.05; 0.25) 0.004**	0.04 (−0.06; 0.14) 0.464	**0.13 (0.02; 0.23) 0.015**
Social sciences	−0.01 (−0.13; 0.10) 0.811	−0.11 (−0.23; 0.01) 0.066	0.04 (−0.08; 0.16) 0.499	−0.06 (−0.18; 0.06) 0.339
Humanities	0.08 (−0.03; 0.19) 0.178	0.05 (−0.06; 0.16) 0.375	−0.03 (−0.14; 0.08) 0.573	0.07 (−0.04; 0.18) 0.242
App food delivery				
No	(ref.)	(ref.)	(ref.)	(ref.)
Rarely	**−0.09 (−0.17; −0.01) 0.026**	**−0.16 (−0.23; −0.08) <0.001**	**−0.15 (−0.23; −0.07) <0.001**	**−0.15 (−0.23; −0.07) <0.001**
At least once/week	**−0.46 (−0.66; −0.26) <0.001**	**−0.38 (−0.58; −0.18) <0.001**	**−0.41 (−0.61; −0.21) <0.001**	**−0.52 (−0.72; −0.32) <0.001**
App food waste				
No	(ref.)	(ref.)	(ref.)	(ref.)
Rarely	0.05 (−0.03; 0.13) 0.181	0.07 (−0.01; 0.15) 0.106	−0.05 (−0.13; 0.03) 0.191	0.06 (−0.02; 0.14) 0.153
At least once/week	**0.30 (0.07; 0.54) 0.011**	**0.50 (0.27; 0.74) <0.001**	0.23 (−0.01; 0.46) 0.059	**0.46 (0.22; 0.70) <0.001**

Cohabitation status results were mixed, with those living with a partner demonstrating a significant positive impact on abilities score ($\beta = 0.23$ (0.06; 0.40), $p = 0.007$) compared to those living alone. However, this variable showed no significant effect on preventive practice or parents' guidance scores. On the other hand, those living with their own family showed a significant increase on the parents' guidance against food waste score ($\beta = 0.18$ (0.04; 0.32), $p = 0.010$) compared to those living alone. Moreover, students with more than four family members exhibited significantly higher parents' guidance scores compared to those with fewer family members ($\beta = 0.18$ (0.10; 0.25), $p < 0.001$).

In terms of academic settings, students in science and technology showed a significant positive association with both abilities scores and total food waste scores than their counterparts in other fields [$\beta = 0.15$ (0.05; 0.25), $p = 0.004$; and $\beta = 0.13$ (0.02; 0.23), $p = 0.015$]. Users of food delivery apps were negatively associated with all assessed food waste categories scores (preventive practices, abilities, parents' guidance) and total food waste scores [$\beta = -0.15$ (-0.23; -0.07), $p < 0.001$ for "rarely" respondents, $\beta = -0.52$ (-0.72; -0.32), $p < 0.001$ for users who more frequently use the app]. On the contrary, frequent users of

anti-food waste apps exhibited a significant positive effect on the preventive practice score, abilities score, and total food waste score [β = 0.46 (0.22; 0.70), $p < 0.001$ for those who use the app at least once in a week].

4. Discussion

4.1. Interpretation of Results

Our study identified several socio-demographic characteristics that significantly influence household food waste behaviors among university students in Italy. These findings provide a comprehensive understanding of how different factors shape food waste practices within this population. However, it is essential to consider the representativeness of our sample when interpreting these results. While the sample includes students from various regions and academic disciplines, the inherent diversity does not necessarily imply that the findings are generalizable to all university students or young adults across Italy or beyond.

The observation that students living outside Lombardy demonstrated slightly higher scores in preventive practices compared to their counterparts in other areas might suggest regional variations in lifestyle or food-related culture. This aligns with research indicating that human behavior and lifestyles can be significantly influenced by differences in community norms, available resources, and local policies [18]. Regional differences in food availability, culinary traditions, and public awareness campaigns might contribute to these variations in food waste behaviors. For instance, Lombardy's more urbanized setting might offer different food access and storage opportunities than rural areas, leading to distinct waste behaviors. However, the sample's composition, with a relatively higher proportion of participants from specific regions, limits our ability to fully extrapolate these findings to other areas of Italy.

Notably, regular smoking and frequent use of food delivery apps were associated with a reduced likelihood of engaging in food waste-preventive behaviors. This finding suggests a potential avenue for targeted interventions that address lifestyle choices as part of broader food waste reduction strategies. Frequent use of food delivery apps may indicate a more "convenience"-oriented lifestyle, which could correlate with less planning and less mindfulness about food consumption and waste. Convenience-driven behaviors, such as ordering takeout, often result in less engagement with meal preparation and food management, potentially leading to increased food waste due to over-ordering or improper storage of leftovers [19]. This pattern of behavior aligns with findings observed during the COVID-19 lockdowns, where convenience and dietary habits, including adherence to the Mediterranean Diet, were significantly impacted [20]. Conversely, larger household sizes and certain academic disciplines (specifically, scientific and technological fields) were linked to a greater propensity for adopting anti-waste behaviors, highlighting the role of community and educational backgrounds in shaping attitudes towards food waste.

The negative correlation between smoking habits and food waste-preventive behaviors requires further exploration. One possible explanation is that smoking, as a health-compromising behavior, may be indicative of lower overall engagement in health-promoting or environmentally sustainable practices. Smokers may be less likely to plan meals or consider the environmental impact of their food choices, resulting in higher food waste [21].

Moreover, previous studies have suggested a link between broader lifestyle patterns and environmental behaviors [22,23]. For instance, research has shown that individuals engaged in health-compromising behaviors, such as smoking, may be less likely to participate in environmentally sustainable actions, such as waste prevention [13], possibly due to differing priorities or levels of awareness [24].

The positive impact of household size on food waste-preventive behaviors corroborates findings from earlier research, which posits that larger families are more efficient in food management, possibly due to the economies of scale in food preparation and the shared responsibility over food resources [25–27]. Among the often-neglected features of healthy diets, such as the Mediterranean Diet, is the preparation and sharing of food.

Furthermore, living with family was associated with stronger parental guidance against food waste, emphasizing the role of family dynamics in fostering environmental stewardship [28]. In the context of the Mediterranean Diet, conviviality and commensality play crucial roles. The preparation and sharing of food within a family setting enhances the efficiency of food management, as family members are more likely to share meals and collaborate in meal planning and preparation. This collective approach not only reduces food waste but also promotes healthier eating habits and strengthens family bonds [29].

Moreover, while our sample included students from diverse academic disciplines, the overrepresentation of students in scientific and technological fields, who displayed a greater propensity for adopting anti-waste behaviors, might have introduced bias. This group's heightened awareness of environmental sustainability through their curricula could disproportionately influence the study's overall results. Although this finding supports the hypothesis that education plays a role in shaping food waste behaviors, it also raises questions about the extent to which this influence applies to students in non-scientific fields [23]. In this perspective, educational programs, therefore, have the potential to tailor their content to enhance not only academic skills but also personal and environmental responsibility. Future research could balance the sample across disciplines to better assess the generalizability of these findings.

Lastly, the divergent effects of app usage on household food waste behaviors are particularly notable. While food delivery apps were associated with pro-food waste behavior [30], the positive impact of anti-food waste apps highlights the potential of technology to support environmental and health goals. This duality underscores the importance of discerning the type of technology used and its potential impacts on behavior.

The literature agrees that household food waste has its roots in human wasteful behaviors [31,32]. Behaviors emerge as the result of how psychological factors interact with the context in which people live [33]. In light of this, three main theories in the literature try to explain how wasteful behaviors are generated. The Theory of Planned Behaviors focuses on psychological aspects: attitude, subjective norms, and perceived behavioral control are the three determinants of intention that, in turn, directly determine behaviors [34]. In this theory, perceived behavioral control also constitutes a direct determinant of behaviors. This means that the stronger the intentions and/or perceived behavioral control to avoid wasteful behaviors, the higher the probability of avoiding those behaviors [10,35].

The Norm Activation Model is another theory based on psychological factors that has two main interpretations [36]. Briefly, in this model personal norms, seen as feelings of moral obligation, are directly linked to behaviors. In this relationship, awareness of the consequences of these behaviors and ascribed responsibility in performing or not performing them can act both as moderators (moderator model) or as determinants of personal norms (mediator model) [37]. Applying this theory to the topic of food waste emphasizes how personal norms are active and can prevent wasteful behaviors when consumers are aware and feel responsible for the negative impacts of food waste [38,39].

Finally, the third construct is the Social Practice Theory that shifts the focus from individuals to the society [40]. In this model, conventions, institutions, infrastructures, and access to resources, or in other terms, the environment, play a fundamental role in determining one's behavior. Practice is viewed as the result of interconnected elements: materials, meaning, and competence. In this perspective, food waste behaviors can be prevented by making available the technologies to reduce food waste, by making those activities that prevent food waste meaningful for individuals and well established in their routines, and finally, by promoting knowledge on practices and reasons to reduce food waste [41–43].

This last model offers a more holistic approach to understanding the deep causes of wasteful behaviors and tends to reflect the complexity of a phenomenon that is multifaceted [44]. Some studies have already investigated the drivers of and barriers to food waste reduction at the household level [45,46]. Including socio-demographic characteristics in these evaluations is essential. In fact, they can help in giving a shape to that environment

that according to the Social Practice Theory is crucial in influencing food waste behaviors. In this light, belonging to older age groups, having a higher educational level, and having a lower income were found to be less likely associated with food waste behaviors [3,47,48]. However, evidence seems controversial and profoundly linked to the setting in which these studies are performed [49].

In summary, our findings reveal that lifestyle choices, such as smoking and the use of food delivery apps, are significantly associated with a lower likelihood of engaging in behaviors that prevent food waste. In contrast, factors such as living in larger households, studying in scientific or technological fields, and the occasional use of food-saving apps correlate with a higher propensity for adopting anti-waste behaviors.

4.2. Implications for Public Health Policies

These findings underscore the importance of incorporating socio-demographic insights into the design of public health policies aimed at reducing food waste. Tailored interventions that consider individual lifestyle factors and leverage educational settings could enhance the effectiveness of food waste reduction campaigns. For instance, initiatives that engage students in scientific and technological fields could harness their propensity for anti-waste behaviors, turning them into advocates for food sustainability within their communities. Educational institutions can integrate food waste reduction topics into their curricula and promote hands-on projects that encourage students to apply their knowledge in real-world settings. Regional differences in food waste behaviors suggest that localized policies and interventions may be more effective than a one-size-fits-all approach. For instance, students living outside Lombardy displayed better preventive practices, indicating that regional cultural norms and policies may influence food waste behaviors. Local governments could develop region-specific strategies that align with the unique cultural and economic contexts of their communities.

The negative association between the use of food delivery apps and food waste-preventive behaviors calls for regulatory measures and educational campaigns to promote more sustainable food consumption practices among users [50]. Public health policies could encourage the incorporation of eco-friendly options within these apps, such as reminders to use leftovers or discounts for ordering smaller portions [51]. Moreover, collaborations with food delivery services to highlight sustainable choices could enhance consumer awareness and reduce food waste [52]. Furthermore, this study highlights the potential of using technology, such as food-saving apps, as a tool for promoting food waste-preventive behaviors. Public policies could support the development and dissemination of such technologies, making them more accessible and appealing to the general public, including university students. By leveraging the growing popularity of apps and digital platforms, policymakers can create innovative solutions that resonate with tech-savvy younger generations.

In conclusion, integrating socio-demographic insights into public health policies and educational programs can significantly enhance efforts to reduce food waste. By addressing the specific needs and behaviors of different population groups, particularly university students, targeted interventions can foster more sustainable food consumption practices and contribute to broader environmental and economic goals.

4.3. Implications for Research

Future research should explore the underlying mechanisms that link socio-demographic characteristics to food waste behaviors, providing deeper insights into how individual and collective factors interact to influence waste practices. Understanding these mechanisms can help identify key factors that contribute to food waste and inform targeted interventions. Moreover, future research should explore these relationships in more diverse populations to understand the generalizability of these findings. Investigating different cultural, economic, and geographical contexts will provide a broader understanding of food waste behaviors and their determinants.

Longitudinal studies could offer a broader understanding of how these behaviors evolve over time, particularly as students' transition from university life to full-time employment. By tracking changes in food waste behaviors, researchers can identify patterns and influences that persist or change over time. Furthermore, since longitudinal studies track changes in food waste behaviors over time, they could help determine the long-term impact of various interventions, such as the introduction of anti-food waste technologies. Such studies would provide valuable data on the sustainability and effectiveness of these interventions, enabling continuous improvement of public health policies [11]. There is also a need to investigate the role of educational programs in shaping food waste behaviors, particularly among students in non-scientific fields. Expanding environmental education across all academic disciplines could foster a broader culture of sustainability [53,54]. Future research could evaluate the impact of integrating food waste education into curricula and assess the long-term benefits of students' behaviors [55]. Lastly, examining the role of family dynamics and household composition in food waste behaviors could provide insights into how family-oriented interventions might be designed. Understanding how family interactions and responsibilities influence food waste can help in developing targeted strategies that address the specific needs of different household types.

In conclusion, addressing food waste requires a comprehensive approach that considers lifestyle choices, technological interventions, educational influences, and family dynamics. By tailoring public health policies to these various factors, it is possible to foster more sustainable food consumption behaviors and contribute to broader environmental and health goals.

4.4. Limitations and Strengths

This study is not without limitations, including its cross-sectional design, which precludes causal inferences, and the self-reported nature of the data, which may be subject to reporting bias or social desirability bias. Future studies could employ more objective measures of food waste and explore causal relationships through experimental designs. Lastly, despite our sample being characterized by a wide range of students from different Italian regions, the representativeness of our study is not nationwide, as universities from various regions of Italy were not included. In this context, future studies could benefit from a multi-center design, which would allow for a more stratified sampling approach that could ensure proportional representation from all regions. Nevertheless, the current study also has important strengths. First, the large sample size of university students is a major strength of the study. By encompassing a wide range of ages, educational levels, disciplinary fields, and living situations, the study provides comprehensive insights into the food waste behaviors of young adults in various socio-demographic contexts. This diversity enhances the generalizability of the findings and ensures that the conclusions drawn are relevant to a broad segment of the university student population. Second, the use of a validated questionnaire to assess household food waste behavior represents another strength of the study. By including items that capture precautionary measures, food management skills, and the influence of parental guidance, the questionnaire offers a detailed and nuanced understanding of behaviors and attitudes related to food waste. This comprehensive approach allows for a more in-depth analysis of the factors influencing food waste beyond what is typically captured in studies focusing on a narrower set of behaviors. Lastly, the application of regression analyses to assess the association between socio-demographic factors and food waste behavior adds statistical rigor to the study. This methodological approach ensures that the findings are robust and reliable, providing a solid foundation for future research and policy development.

5. Conclusions

These findings indicate significant associations between certain socio-demographic characteristics and food waste behaviors among university students in Italy. The observed relationships suggest that both individual lifestyle choices, such as smoking and the use of

food delivery apps, and broader socio-demographic factors, such as household size and field of study, can influence food waste behaviors.

For instance, individual lifestyle choices such as smoking and the use of food delivery apps have been identified as influencing food waste behaviors. Smokers might exhibit different food consumption and waste patterns compared to non-smokers, possibly due to different health awareness levels or time management strategies. Similarly, the increasing reliance on food delivery apps can affect how students plan their meals, manage leftovers, and ultimately, how much food they waste. This reliance might lead to over-ordering or less engagement with traditional meal preparation, resulting in higher amounts of food waste.

Broader socio-demographic factors, such as household size and field of study, also play a crucial role. Students from larger households may have different food waste behaviors compared to those from smaller households, potentially due to shared responsibilities and collective meal planning practices. In larger households, the coordination required to manage food resources might reduce waste, whereas smaller households might experience more individualistic consumption patterns, leading to higher waste.

The field of study is another significant factor. Students in scientific fields may have more exposure to environmental and sustainability issues through their coursework, which can positively influence their food waste behaviors. Conversely, students in non-scientific fields might lack this exposure, leading to less awareness and higher food waste. This highlights the importance of integrating environmental education across all academic disciplines to promote sustainable behaviors universally. These relationships underscore the complexity of food waste behaviors and the need for multifaceted approaches to address them. By understanding how individual and collective factors interact, targeted interventions can be developed. For example, educational programs tailored to different fields of study or campaigns that address the specific needs and habits of smokers and food delivery app users could be more effective in reducing food waste. In summary, the significant associations between socio-demographic characteristics and food waste behaviors among university students in Italy highlight the interplay between individual lifestyle choices and broader socio-demographic factors. These insights can inform the development of targeted interventions and policies aimed at reducing food waste, ultimately contributing to more sustainable consumption patterns and environmental preservation.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/su16219279/s1, Figure S1: Correlation matrix showing the relationships between demographic variables (e.g., age, gender, education) and lifestyle factors (e.g., smoking status, use of food delivery apps, food waste app usage). Positive correlations are shown in blue, while negative correlations are shown in red, with the strength of correlation represented by the intensity of the color; Figure S2: Logistic regression results depicting the association between demographic factors (Region, Education, Status, Family Members, etc.) and three behavioral categories: Preventive Practices, Abilities, and Parental Guidance on Minimizing Food Waste. Each point represents the regression coefficient for a given factor, with the orange markers indicating statistically significant effects. The horizontal lines represent the 95% confidence intervals for each estimate; Figure S3: Logistic regression results illustrating the impact of demographic variables (Region, Education, Status, Family Members, Cohabitation, Smoking Status, etc.) on behavioral outcomes related to food waste management, including app usage (Food Delivery App, Food Waste App). Each point represents the estimated coefficient with corresponding 95% confidence intervals. Significant effects are highlighted in orange.

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